UNIT 2 · TOPIC 2(C)

Springs

Syllabus topic: Design of bearings, couplings, springs and fasteners

Helical compression/extension springs

$$\tau = K_w \cdot 8FC/(\pi d^2)$$

$$K_w = (4C - 1)/(4C - 4) + 0.615/C$$

- Spring index **C = D/d** — recommended **6–12**.
- The Wahl correction factor K_w accounts for curvature + direct shear.

$$\delta = 8FC^3n/(Gd);\ k = F/\delta$$

- Springs in series: **$1/k = \Sigma 1/k_i$**; in parallel: **$k = \Sigma k_i$**.

Other spring types

Torsion springs (trip mechanisms), leaf springs (transport wheels), Belleville washers (clutch loading), rubber springs.

Farm uses

Spring-loaded tines (stony-ground cultivators auto-trip and reset), seed-drill tube pressure springs, clutch springs, relief-valve springs.

Material

Hard-drawn/oil-tempered spring steel (**0.6–0.9% C**); music wire for small springs.

Practice MCQs

Q36. Recommended spring index (C = D/d) for helical springs is:

(a) 1–3 (b) 6–12
(c) 15–25 (d) 30–50

Q37. Wahl factor corrects the stress in helical springs for:

(a) Temperature (b) Curvature and direct shear effects
(c) Buckling (d) Fatigue only

Answers

Q36 — (b) 6–12

Q37 — (b) Curvature and direct shear effects

Sources: Kepner, Bainer & Barger — Principles of Farm Machinery, Ch. 4 (Elements of Rotary Power-Transmission Systems) and Appendix (ASAE V-belt standards); standard machine design references (Shigley; V.B. Bhandari); ASAE/IS standards.